

Introduction to Macroeconomics (S102016CR)

Spring Semester

<https://moodle.unige.ch/course/view.php?id=6618>

Marcelo Olarreaga, Office M5270

```
graph TD; A[Introduction to Macroeconomics] --> B[Part I  
National Accounts and the Real Economy  
(Weeks 1 to 3)]; A --> C[Part II  
Monetary Economics and Financial Economics  
(Weeks 4 to 6)]; A --> D[Part III  
Open Macroeconomics  
(Weeks 7 to 9)]; A --> E[Part IV  
Macroeconomic Policies  
(Weeks 10 to 12)];
```

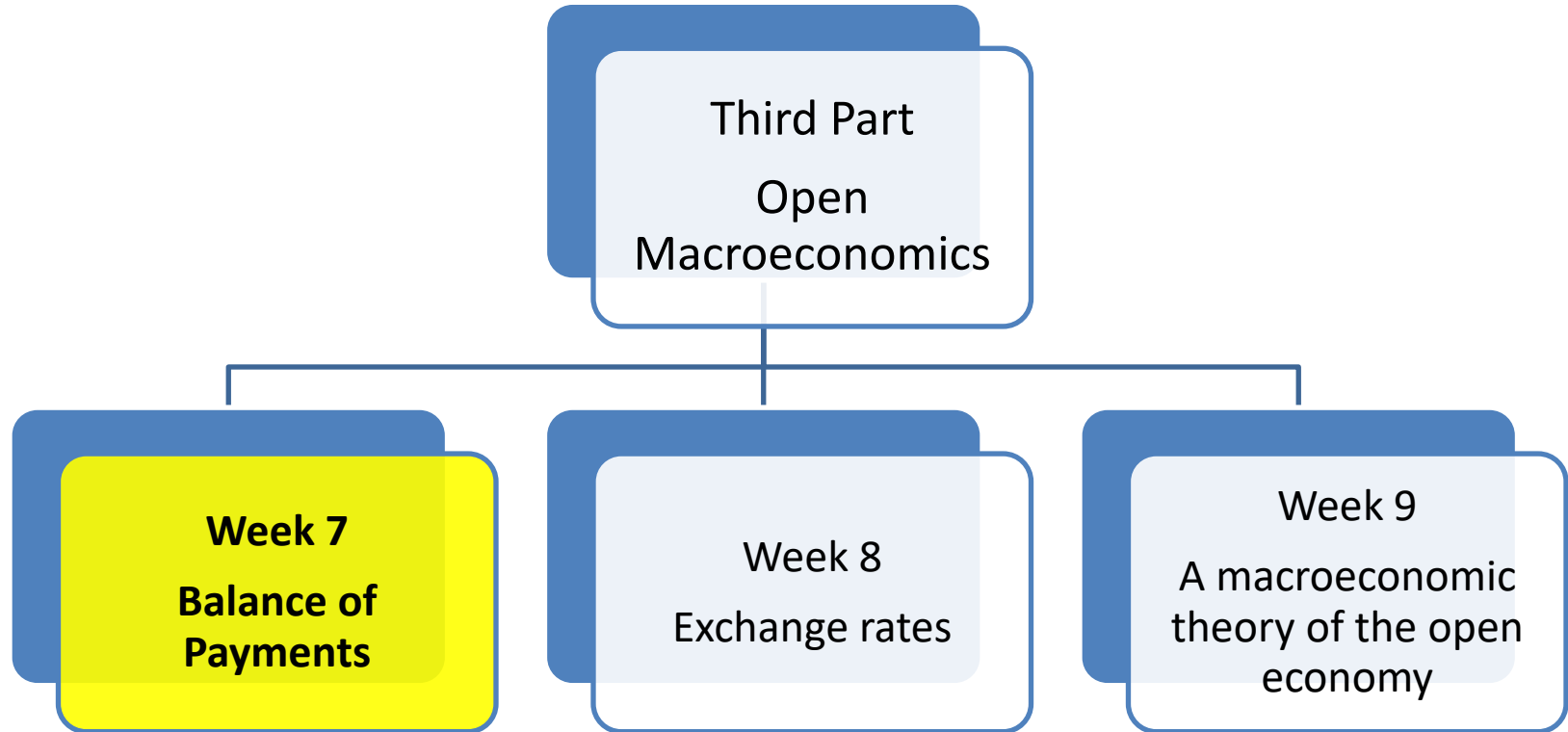
Introduction to
Macroeconomics

Part I
National Accounts and
the Real Economy
(Weeks 1 to 3)

Part II
Monetary Economics and
Financial Economics
(Weeks 4 to 6)

Part III
Open Macroeconomics
(Weeks 7 to 9)

Part IV
Macroeconomic Policies
(Weeks 10 to 12)



Week 7:

Balance of Payments

- In this first chapter on open macroeconomics, we introduce some basic concepts that summarize the economic relationships that intertwine different countries' economies, such as net exports and net capital outflows
- We then define the **current account**, the **financial account** and the **capital account** of the **balance of payments** that help us record all international transactions and discuss how these two relate
- Using national accounting, we then re-explore the relationship between **savings deficit** and **government deficits** but introducing the role of the **trade balance** or **net capital outflows** (i.e., the role of the rest of the world)
- In the following chapters of this third part, we will define exchange rates as the price of foreign currency and explore different models for its determination (jointly with net exports and net capital outflows).

Chapter Contents

- a) Net exports (NX) and net capital outflows (NCO)
- b) The Balance of Payments
- c) Current account and Savings deficits
- d) Summary

Chapter Contents

- a) Net exports (NX) and net capital outflows (NCO)
- b) The Balance of Payments
- c) Current account and Savings deficits
- d) Summary

a. Net exports (NX) and net capital outflows (NCO)

- A *closed economy* does not interact with other economies in the world.
 - There are no exports, no imports, and no capital flows.
- An *open economy* interacts in two ways with other economies:
 - It buys and sells goods and services in world markets (NX)
 - It buys and sells financial and non-financial assets in world asset markets (NCO).
- Let's start with NX. Two components:
 - The *trade balance* is the value of a nation's exports of goods and services minus imports.
 - A *trade deficit* is a situation in which the trade balance is negative.
 - A *trade surplus* is a situation in which the trade balance is positive.
 - *Balanced trade* refers to a case where exports = imports
 - NFIA
- Net exports (NX) or current account is the sum of the trade balance and NFIA.

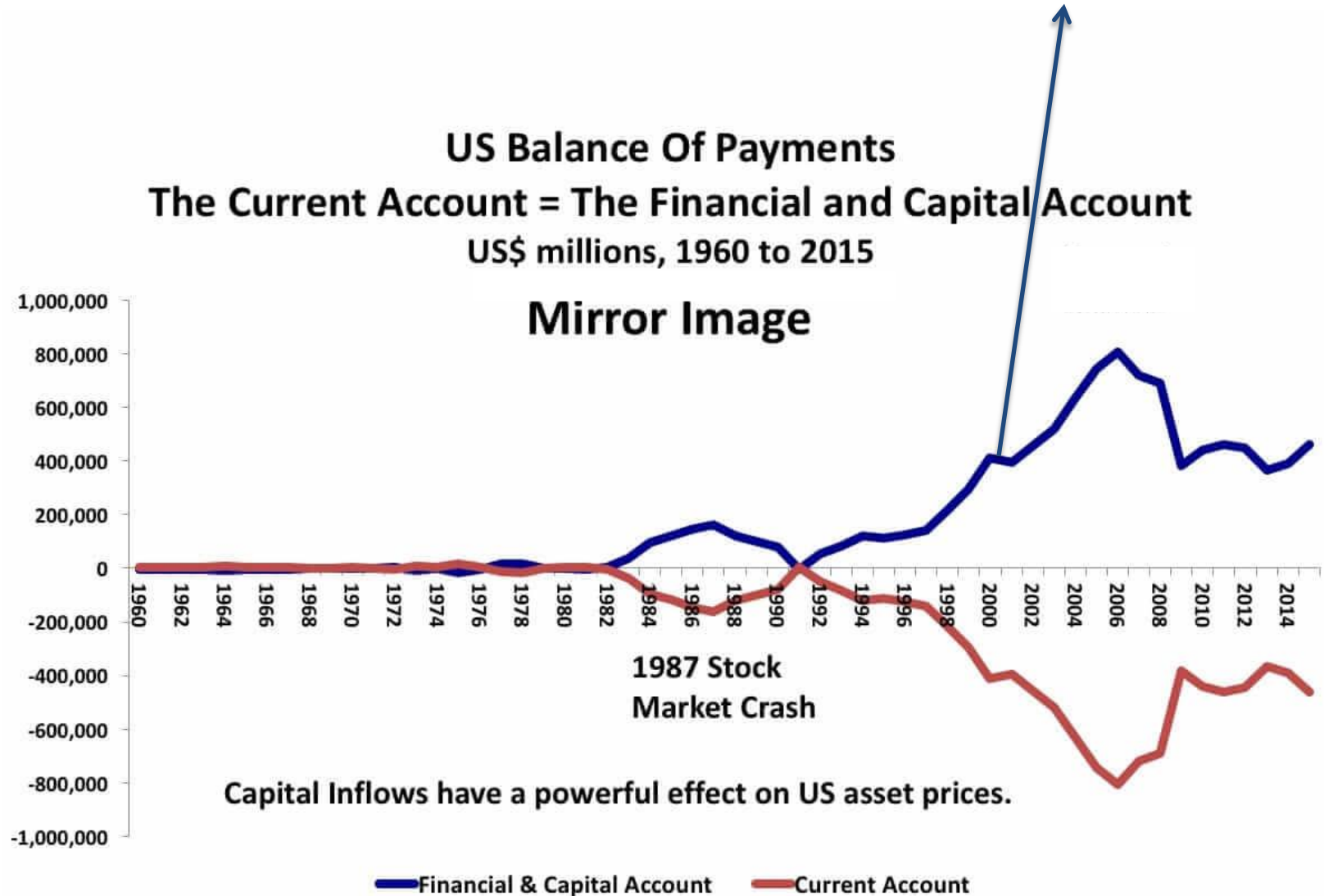
- Factors That Affect Net Exports
 - The tastes of consumers for domestic and foreign goods.
 - The prices of goods at home and abroad.
 - The exchange rates at which people can use the domestic currency to buy foreign currencies.
 - The incomes of consumers at home and abroad.
 - The costs of transporting goods and services from country to country.
 - The government's policies toward international trade, and the temporary movement of persons.

- *Net capital outflows* refer to
 - the purchase of foreign assets by domestic residents minus the purchase of domestic assets by foreign residents
 1. A Chinese resident buys fishing rights from Mauritania's government, increasing China's NCO and reducing Mauritania's NCO
 2. A Swiss resident buys a patent from a German resident, increasing Switzerland's NCO and reducing Germany's NCO
 3. If a Swiss resident builds a new car plant in the UK, this increases Switzerland's NCO and reduces the UK's NCO
 4. If a Swiss resident buys a US government bond, this increases Switzerland's NCO and reduces the US NCO

- Variables that Influence *Net Capital Outflow*
 - The real interest rates being paid on foreign assets.
 - The real interest rates being paid on domestic assets.
 - The perceived economic and political risks of holding assets abroad.
 - The government policies that affect foreign ownership of domestic assets

CA and NCO move together

Financial & Capital account of BoP = NCO



Chapter Contents

a) Net exports (NX) and net capital outflows (NCO)

b) The Balance of Payments

c) Current account and Savings deficits

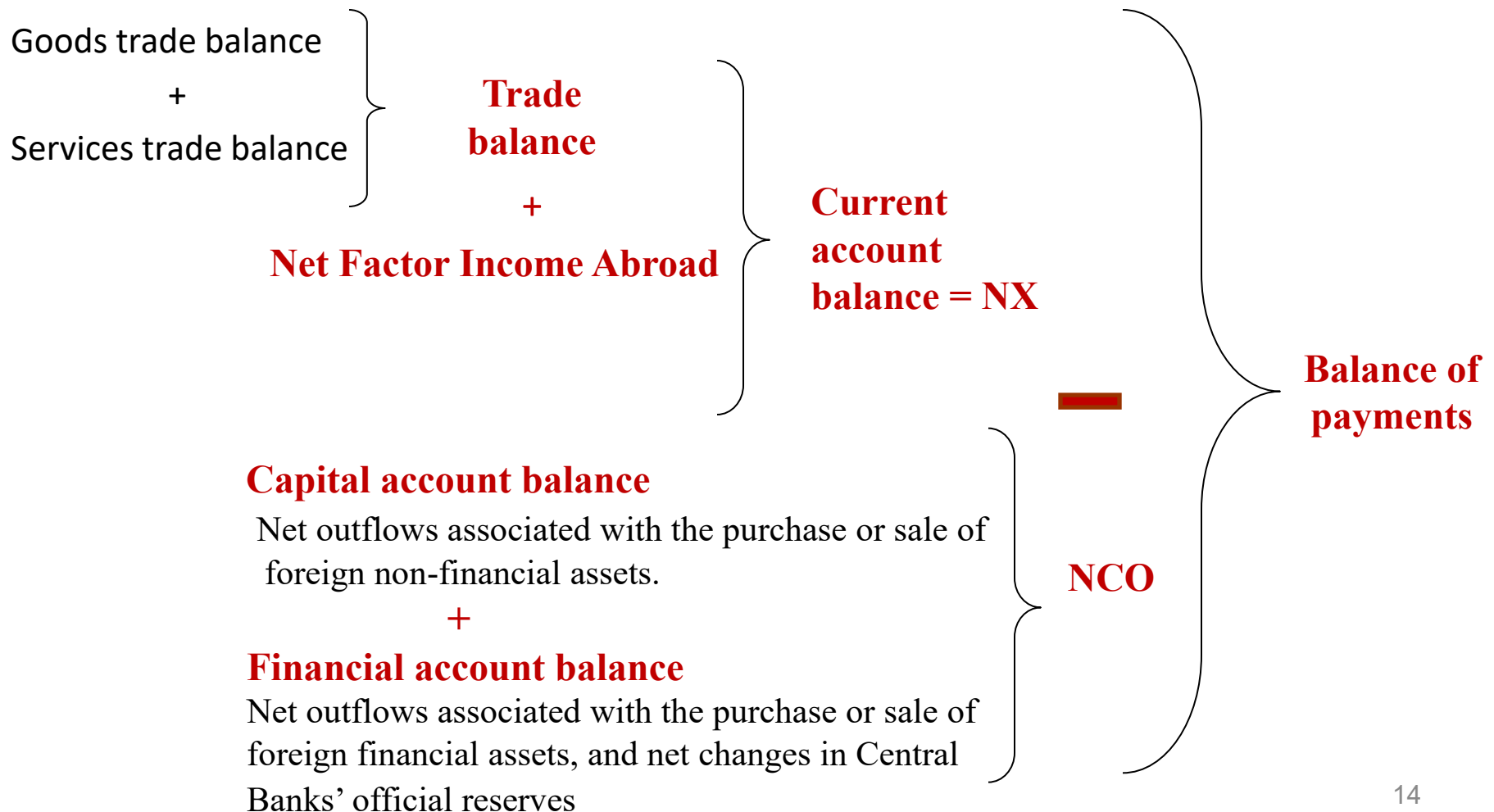
d) Summary

b. The balance of payment

The **balance of payments** (BoP) of a country records all economic transactions between the residents of the country and the rest of the world. The BoP is divided in three parts:

- The Current Account records all the current transactions (export or import of goods and services, residents' earnings from abroad, non-residents' earnings in the country)
= **CURRENT ACCOUNT BALANCE = NX = TRADE BALANCE + NFIA**;
- The Financial Account records the net change in financial outflows with the rest of the world related to financial investment (direct or indirect) or changes in official reserves
= **FINANCIAL ACCOUNT BALANCE = FINANCIAL OUTFLOWS – INFLOWS**
- The Capital Account records the net outflows of capital associated with the purchase or sale of foreign non-financial assets (patents, fishing rights)
= **CAPITAL ACCOUNT BALANCE = NON FINANCIAL ASSETS OUTFLOWS – INFLOWS**
- The sum of the financial account and the capital account equals **NCO (net capital outflows)**

Relationship between the Balance of Payments, the current account (or “net exports” = NX) on the one hand, and the sum of the financial and capital accounts (“net capital outflows” = NCO).



- Thus, net exports (NX) and net capital outflow (NCO) must balance

$$NCO \equiv NX$$

- This holds because every transaction that affects one side must also affect the other by the same amount.

Example: A Fribourg farmer sells Swiss Gruyere to a Boston supermarket

- a. The farmer gives cheese to the supermarket, and the supermarket pays in USD. Swiss exports increased (which raises Swiss NX), and Switzerland has acquired foreign assets in terms of USD (which raises NCO).
- b. The farmer may exchange them for CHF with another entity that wants USD. Two reasons for this (to buy US assets or invest in USD or to buy US goods):
 - i) A Swiss investment fund wants to buy shares in Amazon. In this case, the farmer's net exports equal the net capital outflow to buy Amazon shares.
 - ii) Or another Swiss firm wants to buy Dell computers. In this case, imports will exactly offset export, so net exports (and NCO) are unchanged with respect to the situation before the Swiss farmer exported cheese to the US.

Chapter Contents

- a) Net exports (NX) and net capital outflows (NCO)
- b) The Balance of Payments
- c) Current account and Savings deficits
- d) Summary

c. Current account and Savings deficits

Savings, investment, and their relationship with the NX and NCO

- Current account (CA) is a component of GNP, because $NX = CA = X - M + NFIA$
- $GDP = C + I + G + X - M$
- $GNP = C + I + G + X - M + NFIA = C + I + G + CA = C + I + G + NX = C + I + G + NCO$
- National saving is the income of the nation that is left after paying for current consumption and government purchases:

$$Y - C - G = I + NCO = I + NX$$

- National saving (S) equals $Y - C - G$ so:

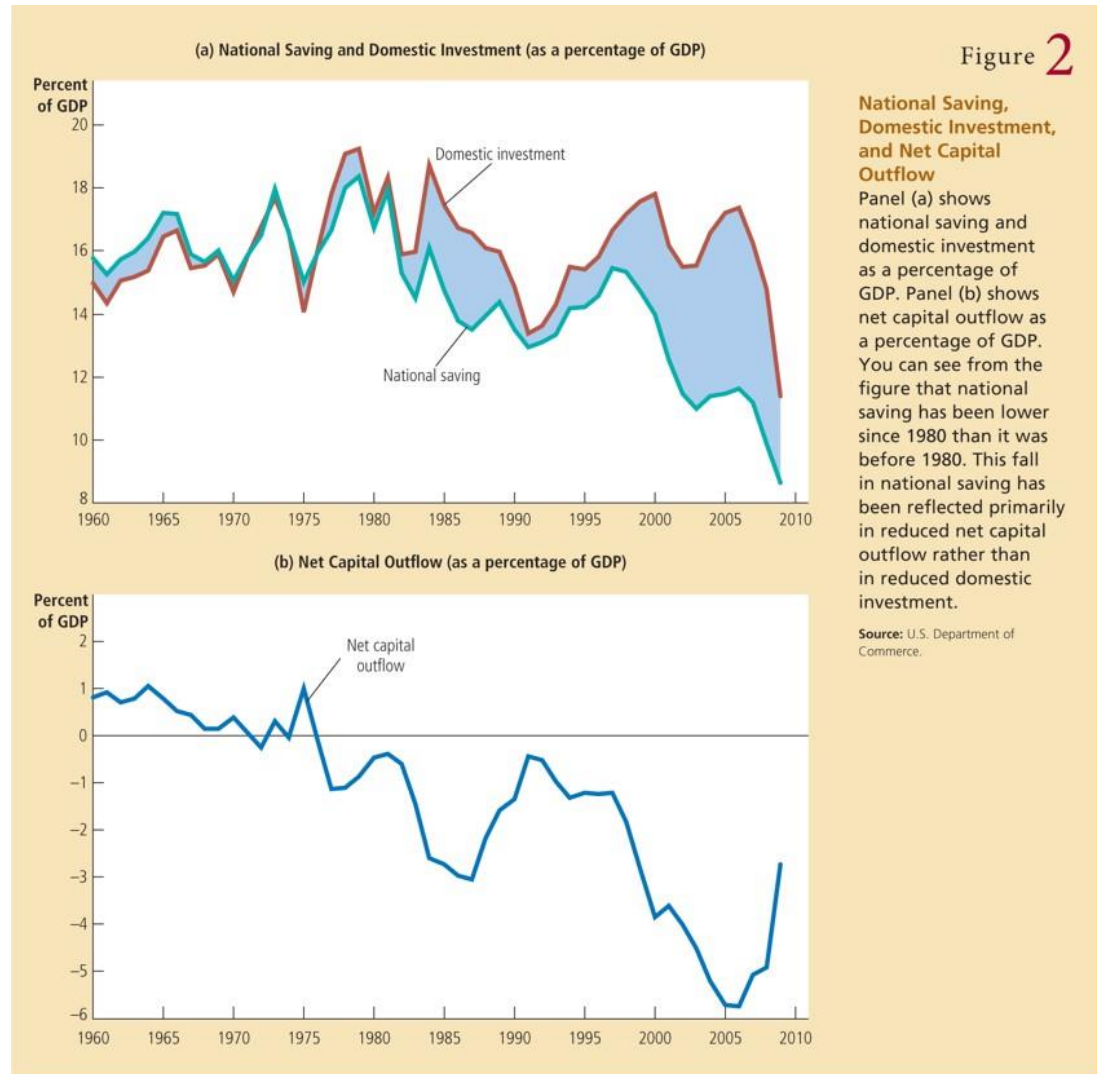
$$S = I + NCO = I + NX$$

Or

$$S - I = NCO = NX$$

- A deterioration of the current account (NX) or NCO is due to a growing savings deficit
- China's trade surplus is explained by a saving surplus

Explaining the negative net capital outflows of the US since 1975: Not enough savings...



- We can also decompose national savings into private and public savings:

$$S - I = NCO$$

$$S^p + (T-G) - I = NCO$$

$$S^p - I = NCO + (G-T)$$

- The savings surplus finances the government deficit and capital outflows

To further understand the relationship between these macroeconomic aggregates, we can also define **domestic absorption** as $C + I + G$

- The difference between domestic production and absorption is equal to what can be lent or needs to be borrowed from the rest of the world:

$$Y = C + I + G + CA$$

$$Y - C - I - G = CA = NCO$$

$$Y - \text{Absorption} = NCO$$

- If $Y > \text{Absorption} \rightarrow NCO > 0$, the country can lend capital to the rest of the world. If $Y < \text{Absorption} \rightarrow NCO < 0$ and the country needs to borrow

Chapter Contents

- a) Net exports (NX) and net capital outflows (NCO)
- b) The Balance of Payments
- c) Current account and Savings deficits
- d) Summary

d. Summary

- Net exports are the value of domestic goods and services sold abroad minus the value of foreign goods and services sold domestically.
- Net capital outflow is the acquisition of foreign assets by domestic residents minus the acquisition of domestic assets by foreigners.
- An economy's net capital outflow always equals its net exports plus net factor income abroad ($NX + NFIA = \text{balance of the current account}$). So, $CA = NCO$
- In the balance of payments, the current account equals the financial account.
- The Balance of payment cannot deteriorate or ameliorate. By definition, it is balanced (apart from measurement error).
- An economy's national savings can be used to finance investments at home or buy assets abroad.